L_0 regularization for subnational microsimulation calibration

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Abstract

Subnational microsimulation requires survey microdata that reproduce administrative totals across nested geographies while remaining usable in a policy model. In the United States, that means calibrating mixed count and dollar targets for district-level units, states, and the nation from a single microdata pipeline. Classical calibration methods provide important reference points, but they do not naturally cover the full production problem: generalized regression (GREG) can produce negative weights and becomes difficult to use in very large, collinear systems, while iterative proportional fitting (IPF, or raking) is most natural for count-style margins.

We present an L_0-regularized calibration pipeline built in PolicyEngine's US data workflow. The pipeline clones CPS households across sampled geographies, constructs a sparse calibration matrix from tax-benefit simulations, and jointly optimizes positive weights and Hard Concrete gates. The gates make sparsity explicit, so the same framework can support compact national datasets and larger subnational datasets. The empirical sections benchmark L_0 against GREG and IPF on shared exported calibration packages, moving from a tractable comparison tier to a scaling frontier and a production-feasibility case. The benchmark reports calibration error, runtime, memory use, failure modes, and retained-record counts.

Keywords: microsimulation, survey calibration, L_0 regularization, subnational analysis

1 Introduction

Tax-benefit microsimulation models apply policy rules to household microdata to estimate fiscal and distributional effects. At the national level, that usually means starting from a survey such as the Current Population Survey (CPS), improving measured incomes and

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program participation, and adjusting weights so that the resulting dataset reproduces known aggregates. Subnational work is harder. Analysts often want estimates for states, congressional districts, or local areas, but the underlying survey was not designed to support all of those geographies at once.

That gap matters in practice. Many reforms are proposed, administered, or debated below the national level. A state tax proposal needs state-specific incomes, filing patterns, and benefit participation. District-level analysis needs those same outcomes to aggregate correctly within each district while remaining consistent with state and national totals. A usable subnational dataset therefore has to satisfy a hierarchical calibration problem rather than a single national one.

Public survey microdata alone do not solve that problem. The CPS provides demographic structure and program coverage, but some tax variables are missing or noisy, top incomes are imperfectly measured, and geographic detail is limited. Administrative targets help, but they arrive from different agencies, with different concepts, and at different geographic levels. In our setting, the calibration problem mixes count and dollar targets across district-level units, the 50 states plus the District of Columbia, and the nation.

The methodological goal is straightforward to state, even if the implementation is not. We want a calibration method that can absorb a large hierarchical target system, keep weights usable for microsimulation, and let us control how large the final dataset becomes. Exact fit is attractive, but a production dataset also needs to remain computationally practical. A file with millions of active cloned records is useful for detailed subnational work and still expensive to simulate. A much smaller file is easier to ship and run, but only if the loss in fidelity is acceptable.

This paper studies an L_0-regularized calibration system built in PolicyEngine's US data pipeline. The system clones CPS households across sampled geographies, builds a sparse calibration matrix from tax-benefit simulations, and jointly optimizes positive weights and binary-like gates. The gates give the optimizer a direct sparsity mechanism. In practical terms, that means the same framework can target a compact national dataset or a larger subnational dataset without switching to a different calibration procedure.

The paper's contribution is empirical as well as methodological. We do not treat generalized regression (GREG) and iterative proportional fitting (IPF, or raking) as universal substitutes for the full production problem. Instead, we use them as classical reference methods within scopes that match their assumptions. The benchmark design uses shared exported calibration packages and asks three questions. First, on a tractable benchmark, does L_0 perform competitively with classical methods when all methods can plausibly run?

Second, how do the methods degrade as the number of targets increases? Third, which methods remain usable near the full production calibration problem?

The rest of the paper follows that structure. Section 2 places the work in the literature on spatial microsimulation, survey calibration, and sparse optimization. Section 3 describes the base microdata and administrative targets. Section 4 presents the calibration pipeline and the benchmark design. Section 5 reports the tractable benchmark, the scaling frontier, and the production-feasibility case. Section 6 interprets those results and discusses their limits.

2 Background and related work

2.1 Subnational microsimulation and spatial reweighting

Subnational microsimulation usually starts from the same basic problem: a national survey contains the behavioural and demographic detail needed for policy modelling, but it does not contain enough observations in every local area to support direct estimation. Spatial microsimulation addresses that gap either by reweighting existing survey records or by constructing synthetic small-area populations from aggregate constraints (Tanton, 2014; O'Donoghue et al., 2014).

This distinction matters for the present paper. Our pipeline remains a reweighting system built on observed CPS households. We do clone households across sampled geographies, but the core empirical problem is still calibration of survey-based microdata to administrative targets. That makes classical calibration methods the closest reference point for the benchmark design.

The spatial microsimulation literature also provides broader context. Williamson et al. (1998), Huang and Williamson (2001), and Harland et al. (2012) describe combinatorial and synthetic-population approaches that search over record assignments area by area. Tanton et al. (2011) and Lovelace and Dumont (2016) discuss reweighting approaches for small-area estimation and policy analysis. These methods are useful precedents, but they usually target one geographic layer at a time. Our setting requires one weighted microdata system to reproduce district-level, state-level, and national targets jointly.

2.2 Survey calibration

Let $i = 1, \ldots, n$ index sampled units, let d_i denote the initial design weight for unit i, and let $\mathbf{x}_i \in \mathbb{R}^p$ denote the vector of auxiliary variables used for calibration. Survey calibration

seeks adjusted weights w_i such that the weighted sample totals match known population totals:

$$\sum_{i=1}^{n} w_i \mathbf{x}_i = \mathbf{T}, \tag{1}$$

where $\mathbf{T} \in \mathbb{R}^p$ is the vector of published totals (Deville and Särndal, 1992; Särndal, 2007). In practice, the auxiliary variables may be counts, indicators, or continuous quantities such as income components. When the target system is internally consistent, classical calibration methods seek exact reproduction of those totals.

The subnational problem in this paper extends that setup in two ways. First, the target vector contains variables from multiple administrative sources and therefore mixes count and dollar constraints. Second, those constraints are nested by geography: district-level totals should add to state totals, which should add to national totals after uprating and reconciliation. In the full production problem, that yields tens of thousands of targets with substantial collinearity across rows.

2.3 Generalized regression

Generalized regression (GREG) calibration minimizes a distance from the initial weights while enforcing the calibration equations (Deville and Särndal, 1992; Särndal, 2007). With the usual chi-squared distance, the calibrated weight for unit i can be written as

$$w_i = d_i + d_i \mathbf{x}_i^\top \left(\sum_{j=1}^{n} d_j \mathbf{x}_j \mathbf{x}_j^\top \right)^{-1} \left(\mathbf{T} - \sum_{j=1}^{n} d_j \mathbf{x}_j \right). \tag{2}$$

GREG is attractive because it can accommodate quantitative targets directly and is well understood in survey statistics.

Its weaknesses are also clear in a microsimulation setting. The matrix in Equation 2 becomes harder to invert reliably as the target system grows and the rows become more collinear. GREG can also assign negative weights. That is acceptable in some estimation settings, but it is awkward in a microdata file that will later be used for tax-benefit simulation, where analysts expect households to represent positive population mass.

2.4 Iterative proportional fitting and raking

Iterative proportional fitting (IPF; Deming and Stephan, 1940; Ireland and Kullback, 1968), often called raking in survey practice, adjusts weights multiplicatively so that selected mar-

gins match published totals. When the margins are compatible and the algorithm converges, the fitted margins are matched exactly. IPF preserves non-negativity and avoids matrix inversion, which makes it a natural baseline for count-style calibration problems.

IPF is narrower than GREG in scope. Classical IPF is built for categorical or margin-style constraints, not arbitrary linear dollar targets. It also does not naturally encode a large mixed system of district, state, and national constraints in one pass. For that reason, the benchmark in this paper uses IPF only on target subsets that fit its assumptions rather than forcing it onto the full production matrix.

2.5 Adjacent weighting literatures

Two adjacent literatures inform the framing even though they are not the main empirical baselines here. Small area estimation borrows strength across areas to improve local estimates from limited samples (Anderson, 2013). Covariate-balancing methods in causal inference choose weights to balance observed covariates between treated and control units (Imai and Ratkovic, 2014). Both literatures share the broader idea that weighting can repair deficiencies in the observed sample, but they optimize for different inferential goals than the hierarchical administrative-target problem studied here.

2.6 L_0 regularization and the Hard Concrete distribution

L_0 regularization penalizes the number of active parameters in a model. In our setting, the parameters are household-geography combinations rather than neural-network weights. Direct optimization of the L_0 norm is combinatorial, so we use the Hard Concrete relaxation proposed by Louizos et al. (2018). Each unit receives a gate $z_i \in [0, 1]$ that is sampled during training and that collapses toward 0 or 1 after clipping. The expected number of active gates is differentiable, which makes it possible to optimize calibration error and sparsity jointly.

That adaptation changes the role of regularization in calibration. Classical calibration methods return one fitted weight vector for the chosen target system. The Hard Concrete gate makes the number of retained records a direct object of optimization. This is the mechanism that lets one pipeline produce both compact national datasets and larger subnational datasets.

2.7 National-level predecessor

Woodruff and Ghenis (2024) developed a national pipeline that imputes tax variables from

the IRS Public Use File onto CPS records and then calibrates the resulting dataset to national administrative totals. The present paper extends that work in three directions. First, the pipeline now assigns households to district-level units and states rather than keeping a single national geography. Second, the optimizer now uses L_0 gates instead of dropout-style regularization, which produces exact sparsity at inference time. Third, the paper evaluates the resulting system as a benchmark study, with explicit comparisons to classical reference methods and an experimental design that tracks what happens as the target system approaches production scale.

3 Data

The calibration pipeline consumes two classes of inputs: household microdata that define the unit universe and administrative targets that define the calibration problem. This section describes both.

3.1 Base microdata

3.1.1 Current Population Survey

The primary microdata source is the Current Population Survey Annual Social and Economic Supplement (CPS ASEC; U.S. Census Bureau, 2024), a nationally representative household survey conducted by the US Census Bureau and the Bureau of Labor Statistics. The CPS provides detailed demographic structure, household relationships, labour income, transfer income, and reported participation in major programmes such as SNAP, Medicaid, SSI, and TANF.

The CPS is not a complete tax-benefit dataset by itself. Some tax variables are missing entirely, others are measured with substantial error, and top incomes are top-coded or otherwise imperfectly captured relative to tax records (Burkhauser et al., 2012). Benefit receipt is also underreported in many surveys relative to administrative sources (Rothbaum and Bee, 2021; Meyer et al., 2021). The pipeline therefore augments the CPS before calibration.

3.1.2 IRS Public Use File imputation

We impute 72 tax variables from the IRS Public Use File (PUF) onto CPS records following the national pipeline of Woodruff and Ghenis (2024). The PUF provides detailed tax-return information but lacks the household structure and demographic coverage needed

for microsimulation. Quantile regression forests (Meinshausen, 2006) bridge that gap by predicting tax variables on the CPS from the PUF training sample.

The imputation proceeds sequentially so that later variables condition on earlier imputed values. This preserves more of the joint tax distribution than an independent-variable approach. The PUF training sample retains the top 0.5% of the AGI distribution in full and samples from the remainder to keep training practical. The resulting enhanced CPS keeps the CPS household structure while importing tax information that is missing or unreliable in the raw survey.

3.1.3 Additional source imputations

The pipeline also fills selected gaps from other surveys:

- The American Community Survey supplies rent and real-estate tax information.
- The Survey of Income and Program Participation supplies tip income and several asset components.
- The Survey of Consumer Finances supplies net worth and selected debt variables.

These source imputations are part of the base microdata construction rather than the calibration step. They define the values that cloned households can contribute to later targets.

3.2 Calibration targets

The target system combines district-level, state-level, and national aggregates from multiple administrative sources. The pipeline stores these targets in a target database, `policy_data.db`. Table 1 summarizes the full active target inventory stored in that database. Individual benchmark runs use subsets of that inventory, so the exact target count in the experiments is reported by benchmark manifest rather than inferred from the database-wide total alone.

3.2.1 District-level targets

District-level targets cover 435 congressional districts plus the District of Columbia. The main sources are:

- IRS Statistics of Income (SOI) targets for AGI, EITC, income components, deductions, and tax-unit counts.

Source	Targets	Geographies
District level		
IRS SOI (AGI, income tax, EITC, 21 income/ded. domains)	25,288	436 district-level units
Census ACS S0101 (person count by 18 age bands)	7,848	436 district-level units
Census ACS S2201 (SNAP household count)	436	436 district-level units
State level		
IRS SOI (same domains as district)	2,958	50 states + DC
Census ACS S0101 (person count by 18 age bands)	918	50 states + DC
USDA FNS SNAP (spending + household count)	102	50 states + DC
CMS Medicaid (enrollment)	51	50 states + DC
Census STC (state income tax collections)	51	50 states + DC
National level		
IRS SOI (domain-constrained aggregates, EITC)	51	1
Curated totals (CBO, JCT, SSA, CMS, Census)	37	1
Census ACS S0101 (person count by 18 age bands)	18	1
Total	**37,758**	

Table 1: Calibration targets by geographic level and source. District-level targets dominate by count. Full variable listing appears in Appendix B.

Source: active targets stored in `policy_data.db`. Individual benchmark runs use subsets of this inventory, so experimental target counts should be reported from the benchmark manifests.

- Census age-distribution targets from ACS tables.
- SNAP household counts derived for district-level use.

The district layer is numerically dominant because each target family is repeated across hundreds of geographic units.

3.2.2 State targets

State targets cover the 50 states plus the District of Columbia. They include:

- IRS SOI targets aggregated to the state level.
- Census age targets.
- USDA SNAP spending and household counts.

- CMS Medicaid enrollment totals.
- Census state income tax collections.

3.2.3 National targets

National targets provide the top layer of the hierarchy. They combine IRS totals with curated aggregates from agencies such as CBO, JCT, SSA, CMS, and Census. Some domains rely on curated published totals stored in the ETL pipeline because there is no single machine-readable source that matches the calibration year exactly.

3.3 Hierarchical uprating and reconciliation

Target sources reference different years and do not always aggregate cleanly across geographic levels. The pipeline applies two adjustments before calibration.

The first is an uprating factor that moves a source-year total to the calibration year. Dollar targets are usually uprated with price or programme-specific growth factors, while count targets use population or enrolment growth where appropriate.

The second is a hierarchy inconsistency factor. For source families where district totals do not sum to the corresponding state totals, the pipeline rescales the district values so that the district layer matches the uprated state layer:

$$\mathrm{HIF}_{s,v} = \frac{T^{\text{state}}_{s,v}}{\sum_{d \in s} T^{\text{district}}_{d,v}}, \tag{3}$$

and

$$T^{\text{adj}}_{d,v} = T^{\text{district}}_{d,v} \times \mathrm{HIF}_{s,v} \times \mathrm{UF}_{s,v}. \tag{4}$$

This reconciliation step does not remove all modelling judgment, but it gives the optimizer a target system that is internally coherent across district, state, and national layers.

4 Methodology

This section first, it describes the calibration system that produces the weighted microdata. Second, it defines the benchmark design used to evaluate that system against classical reference methods.

4.1 Pipeline overview

The pipeline transforms enhanced CPS microdata into a calibrated subnational dataset in four broad steps: cloned geography assignment, calibration-matrix construction, L_0-regularized optimization, and dataset assembly. Figure 1 summarizes that flow.

[Pipeline overview diagram to be inserted]

Figure 1: Pipeline overview. The benchmarked system clones CPS households across sampled geographies, builds a sparse calibration matrix from tax-benefit simulations, optimizes weights with L_0 gates, and then assembles the retained records into deployable datasets.

The benchmark experiments do not rebuild separate pipelines for each method. Instead, they export a shared calibration package from the existing pipeline. That package contains the cloned unit universe, the selected target vector, target metadata, the achievable-target mask, and the initial weights. L_0 and GREG consume that representation directly. IPF uses the same target selection after conversion to an IPF-compatible margin representation.

4.2 Cloned geography assignment

The base CPS contains roughly 100,000 household records. A stratified subsample reduces that to a smaller working dataset while preserving the upper tail of the AGI distribution. This keeps the downstream simulations tractable without collapsing the range of household types the optimizer can use.

Each retained household is then cloned N times, with $N = 430$ in the current production setup. Each clone is assigned to a census block, and the block determines the associated district-level unit, county, tract, and state. The baseline assignment distribution is population weighted. For households above the 90th percentile of AGI, the pipeline tilts that distribution toward high-income districts so that the initial clone universe contains more plausible placements for high-income records:

$$\Pr(B_h = b) = \frac{P_{\text{pop}}(b) R_h(d(b))}{\sum_{b'} P_{\text{pop}}(b') R_h(d(b'))}, \tag{5}$$

where $P_{\text{pop}}(b)$ is the population-based block probability, $d(b)$ is the district-level unit containing block b, and $R_h(d) = A_d$ for high-AGI households and $R_h(d) = 1$ otherwise. Here A_d is the district-level AGI target.

The assignment step also enforces a no-collision rule: the same source household should not appear twice in the same district-level unit across clones unless repeated re-

sampling fails to find an alternative. With 430 clones and the current working dataset, the cloned universe contains roughly 5.2 million household-geography candidates.

4.3 Calibration-matrix construction

The calibration matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$ maps n cloned household candidates to m calibration targets. Entry M_{ji} is the contribution of cloned unit i to target j. For count targets, that is an indicator or count contribution. For dollar targets, it is the simulated value of the relevant tax or benefit variable.

The matrix is built from POLICYENGINE simulations. Because many target variables depend on state-specific policy rules, the pipeline evaluates the cloned households under separate state configurations and reattaches the results according to each clone's assigned geography. For county-specific rules, such as ACA premium tax credits in the current setup, the pipeline can run additional county-level loops. This design keeps the geographic information in the calibration representation rather than in a post-processing step.

Several targets depend on programme take-up. To make the matrix consistent with the final dataset, the pipeline re-randomizes take-up at the clone level using deterministic seeds derived from the source household, the clone index, and the variable name. This lets two clones of the same source household take up a programme at different rates when they are assigned to different places.

The matrix is emitted in sparse coordinate form and stored as a compressed sparse row matrix. After construction, a target-configuration file selects the rows used for a given optimization or benchmark run. Rows with zero feasible support are marked unachievable and excluded from the loss.

4.4 L_0 optimization

Given the sparse matrix $\mathbf{M}$, target vector $\mathbf{t}$, and weight vector $\mathbf{w} \in \mathbb{R}^n$, the optimizer minimizes

$$\mathcal{L}(\mathbf{w}, \boldsymbol{\alpha}) = \frac{1}{m} \sum_{j=1}^{m} \left(\frac{\hat{t}_j - t_j}{|t_j|} \right)^2 + \lambda_{L_0} \sum_{i=1}^{n} \bar{z}_i + \lambda_{L_2} \|\mathbf{w}\|_2^2, \tag{6}$$

where

$$\hat{t}_j = \sum_{i=1}^{n} M_{ji} w_i z_i. \tag{7}$$

The weights are parameterized in log space so that the fitted weights remain positive. The gate z_i is a Hard Concrete random variable with expected activation $\bar{z}_i$, following

Louizos et al. (2018). The first term in Equation 6 is a relative calibration loss, so a 1% error on a small target and a 1% error on a large target receive comparable weight. The second term is the L_0 penalty. It penalizes the expected number of active cloned units and therefore regulates sparsity directly. The third term is an L_2 penalty on the weight vector, that is, a penalty on the squared Euclidean norm $\|\mathbf{w}\|_2^2$. This term shrinks the magnitudes of the fitted weights and discourages solutions that satisfy the targets by assigning very large weights to a small number of retained units.

An L_0 penalty on its own can produce a sparse solution, but sparsity alone does not guarantee a usable microsimulation dataset. The optimizer could still concentrate too much population mass on a small subset of cloned households, yielding unstable or implausible weights even if the support is small. The L_2 term addresses a different problem: it discourages large weight magnitudes conditional on the chosen support. In that sense, the two penalties play complementary roles. λ_{L_0} controls how many units remain active, while λ_{L_2} controls how concentrated the fitted weights can become among those active units.

The separate tuning parameters are therefore a practical advantage of the objective. Increasing λ_{L_0} pushes the solution toward a smaller retained dataset. Increasing λ_{L_2} pushes the solution toward more moderate fitted weights, even if the support size is unchanged. The two knobs let the pipeline trade off calibration fit, sparsity, and weight stability more flexibly than a single regularization term would allow. The full hyperparameter configuration, including the gate parameters and initialization choices, appears in Appendix A. Algorithm 1 gives pseudocode for the full optimization loop.

After optimization, the retained cloned units are assembled into deployable H5 datasets for POLICYENGINE. Those assembly details matter operationally but are not the main object of the benchmark paper, so Appendix D summarizes them.

4.5 Benchmark design

The empirical design is organized around one shared benchmark package and three benchmark tiers. The package is exported from the saved calibration pipeline rather than rebuilt separately for each method. This keeps the comparison focused on calibration method, not on differences in upstream data preparation.

4.5.1 Shared benchmark package

Each benchmark package contains:

- the cloned unit universe

- the selected target vector and target metadata
- the achievable-target mask
- the shared initial weights
- the sparse calibration matrix for scoring fitted weights

GREG uses the exported sparse matrix and target vector directly. IPF requires a conversion step because it works on unit-record tables and margin constraints rather than on a generic sparse linear system. The exporter therefore converts selected count-style targets into an IPF-ready representation while preserving the same cloned unit universe and the same target selection.

4.5.2 Experimental tiers

Tier 1 is a tractable benchmark. It asks whether L_0 performs competitively when the problem is small enough for classical methods to run routinely. The mixed-target tractable benchmark compares L_0 with GREG. A separate count-style tractable benchmark compares L_0 with IPF on a target set that fits IPF's assumptions.

Tier 2 is a scaling frontier. Starting from a controlled benchmark package, we increase the number of targets while holding the basic unit universe fixed. This identifies where each method slows down, destabilizes, or becomes methodologically inappropriate.

Tier 3 is a production-feasibility case. It moves as close as possible to the full production calibration problem, with the current clone count and the current production-style target families. The goal is to study which methods complete, which fail, and document why.

4.5.3 Method scope

The comparison is intentionally asymmetric where the methods are asymmetric. L_0 and GREG can both consume arbitrary linear targets, including mixed count and dollar constraints. IPF is evaluated only on count-style or indicator-style margins. This is a methodological choice rather than a concession. It avoids misrepresenting IPF as a general solver for the full sparse matrix problem.

4.6 Evaluation metrics

Each benchmark run reports:

- mean, median, and maximum absolute relative error across achievable targets
- runtime
- peak memory where available
- convergence or failure status
- any method-specific pathologies, such as negative fitted weights for GREG
- retained-record counts and effective sample size for L_0 outputs

These metrics support the paper's three main empirical questions: comparative performance on tractable problems, degradation as scale increases, and feasibility near production size.

5 Results

The results are organized around the three benchmark tiers defined in Section 4.5. The point of that structure is to separate three questions that the earlier draft treated together: how L_0 behaves on a tractable comparison, how the methods degrade as the target system grows, and which methods remain usable near the production problem. Numerical entries are still marked as placeholders pending completed benchmark runs.

5.1 Tier 1: tractable benchmark comparisons

Table 2 reports the tractable benchmark. The mixed-target setting compares L_0 with GREG on the same exported calibration package. The count-style setting compares L_0 with IPF on an IPF-eligible subset derived from that same package.

The tractable benchmark is the most direct test of whether L_0 performs well only because it was designed for the full production setting. In the mixed-target benchmark, the key comparisons are accuracy, runtime, and whether GREG produces negative weights. In the count-style benchmark, the key question is whether L_0 remains competitive with IPF even when the target system is limited to the class of margins that IPF handles naturally.

5.2 Tier 2: scaling frontier

Table 3 summarizes the scaling ladder. Each rung uses the same basic cloned unit universe while increasing the number of benchmark targets. This isolates how method performance changes with constraint volume rather than with a changing data-preparation pipeline.

Setting		Method	Targets	Mean ARE (%)	Runtime	Notes
Mixed-target benchmark	tractable	L_0	**[TBC]**	**[TBC]**	**[TBC]**	Retained records: **[TBC]**
Mixed-target benchmark	tractable	GREG	**[TBC]**	**[TBC]**	**[TBC]**	Negative weights: **[TBC]**
Count-style benchmark	tractable	L_0	**[TBC]**	**[TBC]**	**[TBC]**	Retained records: **[TBC]**
Count-style benchmark	tractable	IPF	**[TBC]**	**[TBC]**	**[TBC]**	Converged: **[TBC]**

Table 2: Tier 1 tractable benchmark comparisons. The mixed-target benchmark compares L_0 with GREG on the same exported package. The count-style benchmark compares L_0 with IPF on an IPF-eligible subset.

ARE = absolute relative error across achievable targets. Exact target counts come from the benchmark manifests, not the full target database used by the pipeline.

Tier	Targets	L_0 runtime	GREG runtime	IPF runtime	Main outcome
1	250	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
2	500	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
3	1,000	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
4	2,500	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
5	5,000	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
6	10,000	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
7	**[TBC: full selected tier]**	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**

Table 3: Tier 2 scaling frontier. Each row uses the same cloned unit universe and increases the number of selected targets. Runtime cells should be replaced by failure labels when a method does not complete.

The final paper may split this table into separate runtime and failure-mode tables if the notes column becomes too dense.

The scaling frontier is expected to show more than runtime growth. It should also identify where a method becomes numerically unstable, fails to converge, or stops being interpretable as a credible comparison. For GREG, the relevant failure modes include memory pressure, unstable linear algebra, and negative weights. For IPF, the relevant limits include inapplicability outside count-style margins and non-convergence when the margin system becomes too hard to reconcile.

5.3 Tier 3: production feasibility

Table 4 reports the production-feasibility case. This table is designed to be concrete. It should state which method completed, which method failed, how long the run took, and what the failure looked like when a method did not finish.

Method	Target set	Status	Runtime	Peak memory	Notes
L_0	Full production mixed targets	**[TBC]**	**[TBC]**	**[TBC]**	Retained records: **[TBC]**
GREG	Full production mixed targets	**[TBC]**	**[TBC]**	**[TBC]**	Negative weights / solver status: **[TBC]**
IPF	Production count-style subset	**[TBC]**	**[TBC]**	**[TBC]**	Iterations or failure mode: **[TBC]**

Table 4: Tier 3 production-feasibility benchmark. This table should report completion and failure states as directly as possible.

The IPF row is reported on a count-style production subset because IPF is not used as a general solver for the full mixed-target system.

This tier matters because a tractable benchmark alone does not establish practical usefulness. A method can look reasonable at a few hundred targets and still fail to support a deployable subnational dataset. The production-feasibility case links the benchmark back to the actual `policyengine-us-data` use case.

5.4 Supporting diagnostics

The tiered benchmark tables provide the main narrative. The following diagnostics support that narrative by showing how the fitted L_0 solution behaves within a completed run.

5.4.1 Accuracy by geography

Table 5 reports absolute relative error by geographic level for the selected benchmark or production configuration.

Geographic level	Targets	Mean ARE (%)	Median ARE (%)	Max ARE (%)
National	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
State	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
Congressional district	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
All achievable	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**

Table 5: Supporting diagnostic: calibration accuracy by geographic level, measured as absolute relative error (ARE) across achievable targets for the selected benchmark or production run.

Target counts should be read from the exported benchmark package rather than from the full target database used by the pipeline.

The benchmark manifests determine the exact number of targets in each row. Those counts should come from the exported package rather than from the full target database used by the pipeline, which can include multiple years or inactive target families that are not part of a given experiment.

5.4.2 Sparsity and convergence

The sparsity penalty produces exact zeros in the retained weight vector, so dataset size is an explicit output of the optimization rather than a post-processing step. For the settings reported here, the main summary objects are the retained-record count, the effective sample size, and the distribution of fitted weights.

Figure 2 plots mean absolute relative error over training epochs for the selected L_0 runs.

[Convergence curves to be generated from calibration logs]

Figure 2: Convergence diagnostics for representative L_0 runs. The final figure will report the calibration loss over epochs for the tractable benchmark and the production-feasibility run.

These diagnostics are secondary to the benchmark tiers, but they are still important. They show whether a feasible production run is achieving that feasibility by keeping errors low, by keeping weights stable, and by pruning the cloned universe in a controlled way rather than through ad hoc thresholding.

6 Discussion

6.1 What the tractable benchmark establishes

The tractable benchmark is the cleanest answer to the most obvious objection to the method: does L_0 only look good because it is the only method built for the largest problem? That objection is fair, which is why Tier 1 matters. If L_0 performs competitively with GREG on the mixed-target tractable benchmark and remains competitive with IPF on the count-style benchmark, then its later advantages cannot be dismissed as a scale-only story.

This tier should also clarify what kind of advantage L_0 does and does not offer. If GREG matches the tractable targets with similar accuracy but produces negative weights, that is a different kind of limitation from simple inaccuracy. If IPF performs well on count margins but does not extend to mixed dollar-and-count targets, that is a scope limit rather

than a poor implementation. The benchmark design is meant to keep those distinctions visible.

6.2 What the scaling frontier establishes

The scaling frontier shifts the question from local accuracy to method limits. A useful result in this section is not just that one method is faster than another. It is where each method stops being a credible tool for the target system under study. For L_0, the main trade-off is usually between fit, sparsity, and runtime. For GREG, the limits are more likely to come from unstable linear algebra, memory use, or negative weights. For IPF, the limits are more likely to come from scope and convergence.

This frontier is important rhetorically because it distinguishes two claims that are easy to blur. One claim is that L_0 can solve a hard production problem. The stronger claim is that classical reference methods become difficult to apply before the production target volume is reached. The scaling ladder is where that stronger claim lives.

6.3 What the production-feasibility case establishes

The production-feasibility tier connects the statistical benchmark back to deployment. A production run has to do more than minimize loss on a benchmark subset. It has to produce a microdata file that can be published, loaded, and simulated. That is where the sparsity mechanism matters most. Without it, the cloned unit universe remains too large for many downstream uses.

This is also the tier where the paper can be most concrete about failure modes. A table that says a method ran out of memory, returned negative weights, failed to converge, or completed in a given runtime is more informative than a generic statement that the problem is ”large.” The production case therefore should remain a feasibility table first and a performance table second.

6.4 Trade-offs and limitations

The benchmark does not remove all modelling judgment. Several choices remain consequential.

6.4.1 Target-set construction

The exact target count in a run depends on the benchmark manifest, the active target families, and the achievable-target mask. Database-wide counts are useful for context, but the paper should base its empirical claims on the exact exported target set for each run.

6.4.2 Clone count and geography scope

Clone count affects both feasibility and fidelity. More clones enlarge the feasible support for district-level targets, but they also enlarge the optimization problem and the output dataset. The same logic applies to geography scope. Extending the current setup to counties or other local areas is possible in principle, but it would require new target data and a larger benchmark package.

6.4.3 Hyperparameter sensitivity

The L_0 method is not parameter free. The gate temperature, initialization, and sparsity penalty all affect the path the optimizer takes. This is why the hyperparameter configuration belongs in an appendix rather than disappearing into prose. The benchmark should treat those settings as part of the method specification.

6.4.4 Computational cost

The pipeline is computationally heavier than a small classical calibration run. Matrix construction, repeated policy simulation, and gradient-based optimization all cost time. That cost is acceptable only if the pipeline solves a problem that simpler methods do not solve reliably. The benchmark is therefore essential to justifying the added infrastructure.

6.4.5 Comparability of IPF

IPF is the least like-for-like baseline in the paper because its natural target representation is different. Restricting IPF to count-style margins is the right methodological choice, but it also means the reader should not interpret every L_0 versus IPF result as a full-system comparison. The paper should say that directly.

6.5 Generalizability

The specific data engineering in this paper is US-specific, but the benchmark question is broader. Many microsimulation projects need to reconcile survey microdata with admin-

istrative targets across multiple geographic levels. Where those projects also care about keeping the resulting dataset small enough to use, a sparse calibration method may be more useful than a classical exact-fit procedure even if the classical procedure remains attractive on smaller subproblems.

7 Conclusion

This paper studies L_0 regularization as a calibration method for subnational microsimulation. In our setting, the problem is not only to match administrative targets. It is also to do so across nested geographies while producing a dataset that remains usable in an operational policy model.

The paper's main empirical design reflects that goal. Rather than presenting L_0 only as a new optimizer, we benchmark it against classical reference methods on shared exported calibration packages. The tractable benchmark tests whether L_0 is competitive when classical methods can run comfortably. The scaling frontier identifies where those methods begin to fail or fall outside their natural scope. The production-feasibility case connects the comparison back to the `policyengine-us-data` deployment problem.

That framing is important for the substantive claim. A useful subnational calibration method must do more than fit a small benchmark. It must support a large hierarchical target system and produce a manageable microdata file. The L_0 gate is valuable here because it makes sparsity a direct optimization target rather than an afterthought.

The broader contribution is therefore twofold: a sparse calibration method and an open-source pipeline that makes the method deployable. Future work can extend the benchmark to finer geographies, refine the target families, and test alternative sparse optimizers on the same exported calibration packages.

Conflict of interest

The authors declare no conflict of interest.

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Data and code availability

The calibration pipeline, including the L_0 optimizer, is open source at https://github.com/PolicyEngine/policyengine-us-data. The `l0-python` PyTorch package is available at https://github.com/PolicyEngine/l0-python. Calibrated datasets are hosted on HuggingFace at `policyengine/policyengine-us-data`. The IRS Public Use File requires a restricted-access agreement; all other data sources are publicly available.

Use of generative AI

The authors used Claude (Anthropic) for code assistance and manuscript drafting. All technical content, methodology, and analysis were designed and validated by the authors.

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A Optimization hyperparameters

Table 6 lists the default hyperparameters used in the L_0 optimization, together with their code-level names and roles.

Parameter	Code name	Value	Role
Gate temperature	`BETA`	0.35	Controls sharpness of 0/1 gate transition
Left stretch	`GAMMA`	-0.1	Enables exact-zero gates after clipping
Right stretch	`ZETA`	1.1	Enables exact-one gates after clipping
Initial keep probability	`INIT_KEEP_PROB`	0.999	All records start nearly fully active
Weight jitter SD	`LOG_WEIGHT_JITTER_SD`	0.05	Log-space noise on weights at init
Logit jitter SD	`LOG_ALPHA_JITTER_SD`	0.01	Log-space noise on gate logits at init
L_2 penalty	`LAMBDA_L2`	10^{-12}	Mild weight regularization
Learning rate	`LEARNING_RATE`	0.15	Adam optimizer step size
Default epochs	`DEFAULT_EPOCHS`	100	Training iterations (local preset)
Default clones	`DEFAULT_N_CLONES`	430	CPS copies before optimization
λ_{L_0} (local)	`PRESETS[''local'']`	10^{-8}	Sparsity penalty for local-area datasets
λ_{L_0} (national)	`PRESETS[''national'']`	10^{-4}	Sparsity penalty for national dataset
National epochs	(pipeline default)	4,000	Training iterations (national preset)

Table 6: Complete hyperparameter table for L_0 optimization. Source: `unified_calibration.py`.

B Calibration target variables

The calibration pipeline draws targets from seven administrative sources across three geographic levels. The pipeline stores these targets in a target database, `policy_data.db`, which contains 37,758 active targets in total. The following tables list every target domain included in that database, grouped by geographic level.

B.1 District-level targets (33,572)

Target domain	Type	Count
Census ACS S0101		
Person count by age band (18 bands $\times$ 436 district-level units)	count	7,848
IRS SOI		
Person count by AGI bracket (9 bins $\times$ 436 district-level units)	count	3,924
EITC dollars by qualifying children (4 bins $\times$ 436 district-level units)	$	1,744
Tax unit count by qualifying children (4 bins $\times$ 436 district-level units)	count	1,744
Aggregate AGI (unconditional, $\times$ 436 district-level units)	$	436
21 income/deduction dollar totals, each with domain > 0 ($\times$ 436 district-level units)	$	9,156
Tax unit count for each of the 21 domains ($\times$ 436 district-level units)	count	9,156
Census ACS S2201		
SNAP household count ($\times$ 436 district-level units)	count	436
		33,572

Table 7: District-level calibration targets. The 436 district-level units correspond to the 435 congressional districts plus the District of Columbia. IRS SOI provides paired dollar and count targets for each income and deduction domain.

B.2 State targets (4,080)

Target domain	Type	Count
Census ACS S0101		
Person count by age band (18 bands $\times$ 50 states + DC)	count	918
IRS SOI		
Person count by AGI bracket (9 bins $\times$ 50 states + DC)	count	459
EITC dollars by qualifying children (4 bins $\times$ 50 states + DC)	$	204
Tax unit count by qualifying children (4 bins $\times$ 50 states + DC)	count	204
Aggregate AGI (unconditional, $\times$ 50 states + DC)	$	51
20 income/deduction dollar totals (domain > 0, $\times$ 50 states + DC)	$	1,020
Tax unit count for each of the 21 domains ($\times$ 50 states + DC)	count	1,071
USDA FNS SNAP		
SNAP spending ($\times$ 50 states + DC)	$	51
SNAP household count ($\times$ 50 states + DC)	count	51
CMS Medicaid		
Medicaid enrollment ($\times$ 50 states + DC)	count	51
Census STC		
State income tax collections ($\times$ 50 states + DC)	$	51
		4,080

Table 8: State-level calibration targets for the 50 states plus the District of Columbia. IRS SOI variables mirror the district-level structure. USDA provides both SNAP spending and household counts, and CMS provides Medicaid enrollment.

B.3 National targets (106)

Target domain	Type	Count
Demographics (Census ACS + curated)		
Person count by age band (18 bands)	count	18
Person count by SSN card type (4 categories)	count	4
Person count: Medicaid enrolled	count	1
Person count: ACA PTC recipients	count	1
IRS SOI — domain-constrained aggregates		
AGI (unconditional)	$	1
EITC by qualifying children (0–3+)	$	4
Tax unit count by qualifying children (0–3+)	count	4
21 income/deduction dollar totals (domain > 0)	$	21
Tax unit count for each of the 21 domains	count	21
CBO budget projections		
SNAP, Social Security, SSI, unemployment comp., income tax	$	5
SSA benefit decomposition		
Retirement, disability, survivors, dependents	$	4
JCT tax expenditure estimates		
SALT ded., charitable ded., mortgage interest ded., medical expense ded., QBI ded., EITC	$	6
Healthcare spending (MEPS/NHEA/CMS)		
Medicaid, health insurance premiums, Medicare Part B, other medical expenses, OTC health	$	5
Housing, transfers, and other		
Rent, real estate taxes, housing subsidy, work/childcare expenses, TANF, alimony (paid + received), child support (paid + received), tip income, net worth	$	10
Retirement contributions (IRS SOI)		
Traditional IRA, Roth IRA contributions	$	2
		106

Table 9: National-level calibration targets. CBO, JCT, SSA, CMS, and Census values are curated from the cited administrative sources and stored in the ETL pipeline. Dollar values are inflation-adjusted to the calibration year.

C Algorithm pseudocode

Algorithm 1 presents the complete L_0-regularized calibration procedure.

Algorithm 1 L_0-regularized calibration with Hard Concrete gates

Require: Calibration matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$, targets $\mathbf{t} \in \mathbb{R}^m$, initial weights $\mathbf{w}_0 \in \mathbb{R}^n$
Require: Hyperparameters: λ_{L_0}, λ_{L_2}, β, γ, ζ, learning rate η, epochs E
Ensure: Calibrated sparse weight vector $\hat{\mathbf{w}} \in \mathbb{R}^n$

Initialize $\log w_i \leftarrow \log w_{0,i} + \mathcal{N}(0, 0.05^2)$ for all i
Initialize $\log \alpha_i \leftarrow \text{logit}(0.999) + \mathcal{N}(0, 0.01^2)$ for all i
Initialize Adam optimizer with parameters $\{\log w_i, \log \alpha_i\}$ and learning rate η
for epoch $= 1$ to E **do**
    Sample Hard Concrete gates (training):
    for $i = 1$ to n **do**
        $u_i \sim \text{Uniform}(\epsilon, 1 - \epsilon)$
        $\bar{s}_i \leftarrow \sigma\left(\frac{\log u_i - \log(1-u_i) + \log \alpha_i}{\beta}\right)$
        $s_i \leftarrow \bar{s}_i(\zeta - \gamma) + \gamma$
        $z_i \leftarrow \min(1, \max(0, s_i))$
    end for
    $w_i^{\text{eff}} \leftarrow \exp(\log w_i) \cdot z_i$ for all i
    $\hat{t}_j \leftarrow \sum_i M_{ji} \cdot w_i^{\text{eff}}$ for all j
    $\mathcal{L}_{\text{cal}} \leftarrow \frac{1}{m} \sum_{j=1}^{m} \left(\frac{\hat{t}_j - t_j}{|t_j|}\right)^2$
    $\mathcal{L}_{L_0} \leftarrow \sum_{i=1}^{n} \sigma\left(\log \alpha_i - \beta \log \frac{-\gamma}{\zeta}\right)$
    $\mathcal{L}_{L_2} \leftarrow \sum_{i=1}^{n} w_i^2$
    $\mathcal{L} \leftarrow \mathcal{L}_{\text{cal}} + \lambda_{L_0} \mathcal{L}_{L_0} + \lambda_{L_2} \mathcal{L}_{L_2}$
    Backpropagate $\nabla_{\log w, \log \alpha} \mathcal{L}$
    Adam step
end for
Deterministic inference:
for $i = 1$ to n **do**
    $z_i^{\text{det}} \leftarrow \min(1, \max(0,\ \sigma(\log \alpha_i)(\zeta - \gamma) + \gamma))$
    $\hat{w}_i \leftarrow \exp(\log w_i) \cdot z_i^{\text{det}}$
end for
return $\hat{\mathbf{w}}$

D Dataset assembly details

After optimization, the fitted weight vector is reshaped back to clone-by-household form and used to build deployable H5 datasets. The assembly step keeps only active cloned units,

reconstructs the associated person and tax-unit memberships, derives geography from the stored census block assignment, and recomputes geography-dependent quantities such as Supplemental Poverty Measure threshold adjustments.

Two implementation details matter for fidelity. First, geography is derived from the same cloned block assignments that were used to build the calibration matrix, so the assembled dataset matches the unit universe that the optimizer saw. Second, take-up draws are regenerated with the same deterministic seeds used during matrix construction, which keeps take-up-dependent targets consistent between the calibration package and the published dataset.